
EE1060: Differential Equations and Transform Techniques

Lecture 41: Homogeneous Systems of Differential Equations

Topics :

1. Normalized Fundamental Matrix
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Homogeneous System of Differential Equations - Normalized Fundamental Matrix

$$\dot{\vec{x}} = A\vec{x}$$

$\hookrightarrow \in \mathbb{R}^{n \times n}$ (Constant Coefficient matrix, for this course)

fundamental set of sol's = $\{\vec{x}_1, \dots, \vec{x}_n\}$ are of the form

a) $\vec{u}_1 e^{\lambda_1 t} \leftarrow$ distinct eig values. ($\lambda \in \mathbb{R}$)

b) $\left. \begin{array}{l} \vec{u}_1 e^{\lambda_1 t} \\ (t\vec{u}_1 + \vec{u}_2) e^{\lambda_1 t} \\ \vdots \end{array} \right\} \leftarrow$ in rep eig values ($\lambda \in \mathbb{R}$)

$$\vec{u}_j \equiv (A - \lambda_j I) \vec{u}_j = 0.$$

c) $(\vec{\alpha} + j\vec{\beta}) e^{\lambda t} \leftarrow$ complex conjugate pair of λ 's.

Real & imag parts.

solution to any IVP: $\dot{\vec{x}} = A\vec{x}$ with $\vec{x}(t_0) = \vec{x}_0$ is a LC of the fundamental set of solutions.

$$\vec{x} = \sum_{i=1}^n \alpha_i \vec{x}_i \Rightarrow \vec{x}(t) = \vec{\alpha} \quad \text{when} \quad \vec{\alpha} = \begin{bmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{bmatrix}$$

$$\text{Let } \vec{x}(t) = [\vec{x}_1 \dots \vec{x}_n]$$

$\hookrightarrow$ even if $\vec{u} \in \mathbb{R}^n$, $e^{\lambda t}$ makes $\vec{x}$ a function of t .

Normalized Fundamental Matrix - Use

Solution to any Ivp: $\dot{\vec{x}} = A\vec{x}$ with $\vec{x}(t_0) = \vec{x}_0$

$$\vec{x} = X(t)\vec{\alpha} \Rightarrow \vec{\alpha} = X(t_0)^{-1}\vec{x}_0$$

$$\Rightarrow \vec{x} = X(t) \underbrace{X(t_0)^{-1}}_{\vec{\alpha}} \vec{x}_0 \leftarrow \textcircled{1}$$

if $X(t_0) = I \Rightarrow \vec{x} = X(t)\vec{x}_0$

$\hookrightarrow$ for a given t_0 , can we construct $X : X(t_0) = I$.

$\downarrow$

Yes, $X \leftarrow$ Normalized fundamental matrix at t_0 .

Normalized $\rightarrow X_{t_0}(t) \rightarrow \underline{X(t) X(t_0)^{-1}}$
fundamental matrix

Simple case when $\lambda_i \in \mathbb{R}$ and distinct.

fundamental set of solutions = $\{\underbrace{\vec{v}_1}_{\vec{e}_1} e^{\lambda_1 t} \dots \underbrace{\vec{v}_n}_{\vec{e}_n} e^{\lambda_n t}\}$

In order to construct Normalized

fundamental matrix

Scaling doesn't appear to give $\vec{e}_1 \dots \vec{e}_n$

from $\textcircled{1} \quad \vec{x} = \underbrace{X(t) X(t_0)^{-1}}_{\underline{D(t)}} \vec{x}_0$

a) Do all of $D(t)$ constitute a fundamental set?
 $\hookrightarrow$ True.

b) $D(t_0) = ? = I$

The Normalized Fundamental Matrix - e^{At}

for $\dot{\vec{x}} = A\vec{x}$ where $A \in \mathbb{R}^{n \times n}$ (Constant coefficient set of DEs), then e^{At} is a Normalized fundamental matrix at '0'.

Proof:- a) e^{At} is a solution to $\dot{\vec{x}} = A\vec{x}$

$$\text{Let } \vec{x} = e^{At}$$

$$\text{from (2)} \Rightarrow \dot{\vec{x}} = A + \underbrace{At + \frac{t^2}{2}A^2 + \dots}_{A\vec{x}}$$

So e^{At} is a solution

$$b) e^{At} \text{ at } t=0 = I.$$

$$\text{Plug in } t=0 \text{ in (2)} \Rightarrow e^{At} = I \text{ at } \underline{t=0}$$

Normalized $\rightarrow X_0(t) = \underline{e^{At}}$.
fundamental matrix at $t=0$

$\hookrightarrow$ if shown to be true $\Rightarrow \vec{x} = e^{At} \vec{x}_0$
 $\downarrow$
for $\vec{x} = A\vec{x}$ at $\vec{x}(0) = \vec{x}_0$

$$(2) \rightarrow e^{At} = I + At + \frac{t^2}{2}A^2 + \frac{t^3}{3!}A^3 + \dots +$$

$$\text{if } A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \rightarrow e^{At} = \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix}$$

$$\text{if } A = \begin{bmatrix} \end{bmatrix}$$

$$A = [D_{\text{diag}}] + LT + UT.$$